

Evidence Brief | Spotlight Series No. 3

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Climate change is increasingly straining Africa's health systems, yet financing for climate-health action remains heavily dependent on traditional donors. As climate shocks disrupt health services, infrastructure, and supply chains, there is an urgent need to explore sustainable, locally driven financing mechanisms that can strengthen health system resilience beyond conventional donor funding.

Why Donor Money Alone Won't Be Enough.

For years, climate and health work in Africa has leaned heavily on donor funding. That funding is now shrinking, and it is not coming back to previous levels. At the same time, the gap is large: Africa needs roughly USD 70 billion a year by 2030 to adapt to climate change, and currently the private sector provides only a small slice of that money. In Tanzania, for example, private sectors make up about 5% of climate finance. This financing gap means African governments cannot rely on donor funding alone and will need to mobilize more domestic resources while creating greater opportunities for private sector investment. To achieve this, four key shifts are needed: financing should be blended from multiple sources rather than relying on a single donor; health solutions should be designed to become financially sustainable; local businesses should play a larger role; and partnerships should focus on delivering measurable results rather than existing only on paper.

What the Private Sector Brings to the Table

The private sector brings unique comparative advantages to the climate-health nexus by providing the capital, innovation, technology, infrastructure, and operational scale needed to accelerate resilient health systems. Leveraging expertise in manufacturing, supply chains, logistics, and digital solutions, private companies can rapidly deploy and scale climate-smart interventions across multiple countries. These strengths translate into practical contributions such as powering health facilities with renewable energy, developing climate-resilient diagnostics, expanding digital health and early warning systems, strengthening local manufacturing and cold-chain logistics, and mobilizing innovative financing to support integrated climate and health investments.

Four Building Blocks for Climate Resilient Health System

To operationalize private sector engagement in the climate-health nexus, four key building blocks are essential. The first priority is mobilizing private sector investment and innovation to build resilient health infrastructure through intelligent, flood-resistant facility designs and reliable backup systems that can sustain operations during extreme heat and other climate-related shocks. The second is renewable energy, with a focus on harnessing Africa's vast solar potential to power health facilities and reduce dependence on diesel and other fossil fuels.

The third building block is green supply chains, encompassing climate-resilient logistics that emissions through route optimization, low-carbon transport solutions such as electric drone delivery where appropriate, and energy-efficient vaccine cold chains. The fourth is digital health and climate-health data integration, encompassing surveillance systems and remote diagnostics that can improve service delivery while reducing the climate burden on health systems. Collectively, these demonstrate that climate resilience must be integrated across every component of the health system to ensure sustainable, reliable and effective healthcare delivery.

Evidence From the Field

These building blocks are already being applied through several practical examples. In Nigeria, EM-ONE Energy's solar-plus-storage microgrids power 90 health facilities. In Ethiopia, a solar-and-battery hybrid system at St. Luke's Hospital sustains operations during power outages. In Rwanda, mRNA vaccine manufacturing nearing completion is being powered by a sustainable, equitable renewable energy system. Zipline's drone-based delivery of blood, vaccines, and medicines now operates across Rwanda, Ghana, Nigeria, and Kisumu, Kenya, strengthening climate-resilient last-mile healthcare delivery.

On cold-chain equipment, Gavi has deployed over 15,300 pre-qualified, solar-direct-drive off-grid vaccine refrigerator units across Africa. In Uganda, Rocket Health's digital platform enables remote consultation and lab sample pickup, reducing the need for patient travel and its associated environmental impact. Africa Health Business itself has piloted Black Frog, a solar-powered, back-portable cold-chain device that maintains vaccine temperature through an integrated, phone-linked SMS recording system, offering a replicable, affordable, low-carbon model for last-mile vaccine delivery already in use across several Kenyan counties and ready for scale-up.

New Pathways for Financing Climate-Health

Climate and health financing must shift from donor dependence to sustainable investment models that position climate resilience as an economic opportunity. Governments should leverage blended finance, green bonds, corporate Environmental, Social, and Governance (ESG) investments, outcome-based financing, and public-private partnerships to mobilize domestic and private capital for climate-resilient health systems.

A major barrier to attracting investment is the limited availability of investment-grade evidence. Strengthening economic analyses, including the costs of climate-related illness, health system disruptions, and climate-attributable disease projections, will help demonstrate returns on investment, reduce perceived risks, and unlock financing from domestic and private investors.

What this Means for Decision Makers

To enable this transition, policymakers should create an environment that incentivizes private sector participation through supportive regulations, innovative financing mechanisms, and stronger links between evidence and investment decisions. Financing should prioritize long-term resilience by supporting local manufacturing, renewable energy, digital health, resilient supply chains, and climate-informed health services. Sustainable climate-health financing will ultimately depend on locally led, community-centred investments that generate lasting economic and health benefits beyond traditional donor funding.